

Pythagoras' Theorem & Trigonometry

Extra practice test 2 · 27 marks total

Name: _____ Date: _____ Score: _____ / 27

Section A — Pythagoras' theorem

A1. A right-angled triangle has shorter sides of 9 cm and 12 cm. Find the length of the hypotenuse.

[2 marks]

A2. A right-angled triangle has a hypotenuse of 17 cm and one shorter side of 8 cm. Find the length of the other shorter side.

[3 marks]

A3. A rectangle measures 9 cm by 40 cm. Find the length of its diagonal.

[2 marks]

Section B — SOH CAH TOA basics

B1. In a right-angled triangle, what is the name of the side opposite the right angle?

[1 mark]

B2. Write down the SOH CAH TOA formula you would use to find the adjacent side if you know the hypotenuse and the angle.

[1 mark]

B3. Write down the SOH CAH TOA formula you would use to find an angle if you know the opposite and hypotenuse.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has an angle of 37° and a hypotenuse of 12 cm. Find the length of the side opposite the angle, to 1 decimal place.

[3 marks]

C2. A right-angled triangle has an angle of 62° and an adjacent side of 5 cm. Find the length of the hypotenuse, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 27° and an adjacent side of 9 cm. Find the length of the side opposite the angle, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A right-angled triangle has an opposite side of 8 cm and a hypotenuse of 17 cm. Find the angle, to 1 decimal place.

[2 marks]

D2. A right-angled triangle has an adjacent side of 7 cm and an opposite side of 11 cm. Find the angle, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A ladder rests against a vertical wall. The foot of the ladder is 2.1 m from the wall and the ladder is 5.8 m long. Calculate how far up the wall the ladder reaches, to 1 decimal place.

[2 marks]

E2. A wire runs from the top of a vertical pole to a point on the ground 8 m from its base, making an angle of 50° with the ground. Calculate the height of the pole, to 1 decimal place.

[3 marks]